Computational Études

A Spectral Approach

A Tentative Teaching Plan

Course Information

Course: MATH 794 – 01 (CRN 21296)
Semester: Spring 2026
Schedule: Mon & Wed, 16:30 – 17:45
Location: G-G02013

Instructor

Name: Dr. Denys Dutykh
Department: Mathematics
Institution: Khalifa University
Location: Abu Dhabi, UAE

Legend: Completed Current Examination Planned

Week	Dates	Monday Lecture	Wednesday Lecture
1	Jan 12–14	Course introduction Syllabus & overview	**Ch. 2:** Classical PDEs Heat equation – separation of variables
2	Jan 19–21	**Ch. 3:** Mise en bouche Method of weighted residuals, collocation example, collocation vs Galerkin – Project I assigned	**Ch. 4:** The Geometry of Nodes Lagrange interpolation, Runge phenomenon, potential theory
3	Jan 26–28	**Ch. 5:** Differentiation Matrices FD stencils, spectral matrices, Fornberg algorithm	**Ch. 6:** Smoothness and Spectral Accuracy Smoothness, spectral convergence, exponential accuracy
4	Feb 2–4	**Ch. 7:** Chebyshev Differentiation Matrices Chebyshev nodes, D_N matrix, negative sum trick	**Ch. 8:** Boundary Value Problems 1D/2D BVPs, matrix stripping, Newton iteration – Project II assigned
5	Feb 9–11	**Project I:** Students Presentations Oral presentations and discussions of Project I results	**Ch. 9:** Fourier Space on Grids Fourier transform, DFT/FFT, aliasing, sinc interpolation

Week	Dates	Monday Lecture	Wednesday Lecture
6	Feb 16–18	**Ch. 10:** Spectral PDE Solvers Chebyshev–Fourier recap, Chebyshev differentiation via FFT, method of lines	**Ch. 11:** Pseudo-spectral PDE Solvers Fourier Pseudospectral Methods – Project III assigned
7	Feb 23–25	**Ch. 12:** Spectral Methods in Polar Coordinates Coordinate singularity, Fornberg's doubling trick, Chebyshev–Fourier disk Laplacian, Bessel eigenmodes	**Ch. 13:** Advanced Boundary Conditions Lifting (Method I) vs tau (Method II), Robin/Neumann BCs, nonlinear BCs, QEP and quasinormal modes
8	Mar 2–4	**Project II:** Students Presentations Oral presentations and discussions of Project II results	**Case Study** assigned Replaces Midterm Examination
–	Mar 9–22	*Spring Break & Eid Al Fitr – No classes*	
9	Mar 23–25	**Ch. 14:** Higher-Order BVPs Polynomial trick for clamped BCs, $u^{(4)} = e^x$ (Étude 14.1), beam eigenmodes (Étude 14.2)	**Project III:** Students Presentations Oral presentations and discussions of Project III results
10	Mar 30–Apr 1	Spectral Method Convergence for QNMs (I) Overview of the convergence proof for spectral methods applied to quasinormal modes – Slides: `slides/QNM-SpectralConvergence.pdf`	Spectral Method Convergence for QNMs (II) Continuation and discussion of the convergence proof – Project IV assigned
11	Apr 6–8	**Ch. 15:** Quadrature in Spectral Methods Exactness principle, Newton–Cotes failure, Clenshaw–Curtis vs Gauss, aliasing in Chebyshev space	**Ch. 16:** Periodic Quadrature Trigonometric exactness, five convergence classes, Poisson's ellipse, trapezoidal rule on the real line
12	Apr 13–15	**Chapter 17:** Time Stepping, Stability, and the CFL Constraint Method of lines, stability regions, Fourier and Chebyshev CFL conditions, overcoming the CFL barrier	**Case Study:** Oral Presentations Presentations and report submissions – Wed, Apr 15
13	Apr 20–22	**Chapter 18:** Linear Spectral Eigenproblems Discretisation, drift-with-N diagnostic, numerically vs physically spurious eigenvalues, the complex-plane detour, the trust certificate	**Chapter 19:** Coordinate Transformations and Mapped Spectral Methods Chebyshev-as-cosine, general mapped calculus, weak-endpoint-singularity cure via tanh, arctan/tan map for periodic concentration, Kosloff–Tal-Ezer trade-off

Week	Dates	Monday Lecture	Wednesday Lecture
14	Apr 27–29	**Project IV:** Oral Presentations Presentations and report submissions – Mon, Apr 27	**Chapter 20:** Spectral Methods on Unbounded Intervals Domain truncation, sinc / Hermite / Laguerre bases, rational-Chebyshev TB_n and TL_n, behavioural BC at infinity, oscillatory-tail augmentation
–	May 4–14	**Final Examinations Period** **Final Exam: Thursday, May 7, 2026**	

Table 1: Teaching schedule for MATH 794 – Spring 2026

Notes

- This schedule is tentative and may be adjusted based on class progress.
- Chapter numbers refer to *Computational Études: A Spectral Approach*.
- Office hours are available by appointment.
- All course materials are available in the course repository.

Course Projects

Project I: Spectral Methods for Boundary Value Problems (Based on Chapter 3 – Assigned: Jan 19, 2026)

Implement spectral methods (collocation and/or Galerkin) to solve a boundary value problem of your choice. Requirements:

1. Find a linear BVP of at least second order with **non-homogeneous** boundary conditions.
2. Construct an exact solution to this problem (or design the problem around a known exact solution).
3. Choose appropriate basis functions that satisfy the boundary conditions identically.
4. Apply the **Collocation** method, **Galerkin** method, or both to obtain a numerical solution.
   - If using **only** the Collocation method, the BVP must have **non-constant coefficients**.
   - If applying **both** methods, the problem may have constant coefficients (though non-constant coefficients are also welcome).
5. Present results in a table that includes the error at the collocation points.
6. Provide graphical representations showing: the exact solution, the numerical approximation, and the difference (error) between them.

Deliverables: A written report, a short oral presentation (15 min max), and a live demonstration of your computational codes. Grading is based on relative performance within the class.

Project II: Spectral Collocation for BVPs (Based on Chapters 6–8 – Assigned: Feb 2, 2026)

Implement Chebyshev spectral collocation to solve boundary value problems. Requirements:

1. Implement the Chebyshev differentiation matrix D_N using the negative sum trick.
2. Choose a second-order BVP (linear or nonlinear) with known exact solution.
3. Solve the BVP using spectral collocation with matrix stripping for boundary conditions.
4. Present a convergence study: error vs. N on a semilog plot.
5. For bonus: solve a 2D problem using tensor products.

Deliverables: A written report, a short oral presentation (15 min max), and a live demonstration of your computational codes. Grading is based on relative performance within the class.

Project III: Fourier Pseudospectral Simulation of Periodic PDEs (Based on Chapters 9–11 – Assigned: Feb 18, 2026)

Implement a Fourier pseudospectral method to simulate the dynamics of a PDE of your choice on a periodic domain. Requirements:

1. Choose a time-dependent PDE in (x, t) variables posed on a periodic spatial domain.
   - The PDE may be linear (e.g. advection, diffusion, Schrödinger) or nonlinear (e.g. Korteweg–de Vries, Burgers, Allen–Cahn, Kuramoto–Sivashinsky).
2. Discretize spatial derivatives using the Fourier pseudospectral method (via the FFT).
3. Advance in time using an appropriate ODE solver (e.g. Runge–Kutta, exponential integrator, or split-step method).

4. Validate your implementation against an exact solution, a conserved quantity, or a known qualitative behaviour of the chosen PDE.
5. Present space-time visualizations of the computed dynamics (e.g. waterfall plots, animations, or snapshots at selected times).
6. For bonus: extend the simulation to two or three spatial dimensions.

Deliverables: A written report, a short oral presentation (15 min max), and a live demonstration of your computational codes. Grading is based on relative performance within the class.

Project IV: Open Topic in Spectral Methods (Based on Chapters 12–14 – Assigned: Apr 1, 2026)

Choose any topic from the textbook *Computational Étude: A Spectral Approach*, starting from Chapter 12 and until the end of the book. The choice of topic is left to each student according to their personal taste and scientific preferences. Requirements:

1. Select a problem, method, or application from Chapters 12–14 of the textbook that interests you.
   - Examples include (but are not limited to): spectral methods in polar coordinates, eigenmodes on the disk, advanced boundary conditions (Robin, radiation, quasinormal modes), higher-order BVPs, the biharmonic operator, Orr–Sommerfeld stability, pseudospectra, or the Kuramoto–Sivashinsky equation.
2. Implement the chosen method or solve the chosen problem computationally, in any of the three programming languages adopted in our course (Python, MATLAB, or Julia).
3. Validate your implementation against known analytical results, reference solutions, or established benchmarks from the literature.
4. Present your findings through appropriate visualizations (convergence plots, mode shapes, space-time diagrams, *etc.*).
5. Discuss the strengths and limitations of the spectral approach for your chosen problem.

Deliverables: A written report, a short oral presentation (15 min max), and a live demonstration of your computational codes. Grading is based on relative performance within the class.

Last updated: April 24, 2026